United International University

Department of Computer Science and Engineering

CSE 2213/CSI 219: Discrete Mathematics (BSCSE and BSDS)

Mid-term Examination – Summer 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. (a) Consider the following propositions: [1*2=2]

   p: Today is Sunday.
   q: I do homework.
   r: I watch TV.

   Express the following statements using propositions and logical connectors:

   i. I do homework every day except Sunday.
   ii. If today is Sunday, then I watch TV; otherwise I do homework.

   (b) Determine the contrapositive and converse of the following statement: [2]

   "Barcelona cannot win the match unless Lamin Yamal plays."

   (c) Using logical equivalence laws, prove that the following expression is a tautology: $\neg(p \to q) \to \neg q$ [2]

2. (a) Consider the following predicates: [1*3=3]

   $S(x)$: x is a student
   $B(x)$: x has a book
   $L(x, y)$: x likes y

   The domain consists of all people. Express the following statements using quantifiers, logical connectives, and the predicates above:

   (i) All students have a book.
   (ii) There is someone who is liked by everyone.
   (iii) Every student either has a book or likes someone who has a book.

   (b) Determine whether the following propositions are true or false. Justify your answer. The domain of all variables is the set of real numbers. [1*3=3]

   (i) $\forall x \exists y \forall z\ (x + y = z)$
   (ii) $(\forall x(x^2 \geq 0)) \leftrightarrow (\exists y(y^2 < 0))$
   (iii) $\exists x \forall y\ ((x \cdot y = x) \to (x = 0 \lor y = 1))$

3. Let's assume: $A = \{1, 2, 3, 4, 5\}, B = \{5, 2, 1, 0, 9\}, C = \{0, 1, 7, 5, 8\}$ and $D = \{0, 6, 7, 8, 9\}$.

   (a) Express set A in Set Builder Method. [1]
   (b) Find the power set of $(C - D)$ [1]
   (c) Find the set $(A \cup D) - (B \cap C)$ [2]
   (d) Find the cardinality of $(A \cap B) \times (B \cap C \cap D)$ [2]

4. (a) Determine whether each of the following is a function. Justify your answer. [1*2=2]

   i. $f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \dfrac{2x + 3}{6x - 2}$
   ii. $g : \mathbb{R} \to \mathbb{R}, \quad g(x) = \pm\sqrt{x}$

   (b) Show that the following function is invertible. Find its inverse function $f^{-1}(x)$, and evaluate $f^{-1}(2)$. [4]

   $$f : \mathbb{R} - \left\{\frac{1}{2}\right\} \to \mathbb{R} - \{1\}, \quad f(x) = \frac{x + 1}{2x - 1}$$

5. (a) Prove the following statement using an appropriate proof technique: **"If r is irrational, then $\sqrt{r}$ is irrational."** [3]

   (b) Prove the following statement using an appropriate proof technique: **"For all integer n, if $5n^2 - 6n + 7$ is even, then n is odd"** [3]